



Problem Set 8

Due: 19 July 2024, 3:30 p.m.

Problem 1. Based on measurements of rotational energy levels of a HF molecule, its moment of inertia about the axis perpendicular to the line connecting both atoms, passing through the center of mass of the system, has been estimated at $1,37 \cdot 10^{-47} \text{ kg} \cdot \text{m}^2$. Find the distance between the H and the F atoms if their masses are $m_{\text{H}} = 1,67 \cdot 10^{-27} \text{ kg}$, $m_{\text{F}} = 3,17 \cdot 10^{-26} \text{ kg}$, respectively.

(2 points)

Problem 2. Justify the following statement that we mentioned in class when we were discussing the moment of inertia of a disk: *For a planar object, the moment of inertia about an axis perpendicular to the plane is the sum of the moments of inertia about two perpendicular axes through the same point in the plane of the object.*

It is known as *the perpendicular axis theorem*.

(3 points)

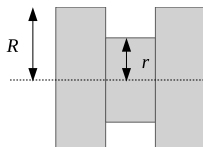
Problem 3. Find the moment of inertia of a hollow cylinder with inner radius R_1 , outer radius R_2 , and height H , about the axis of symmetry. The cylinder is not uniform and the bulk density of mass depends on the distance r from the axis of symmetry as αr with $\alpha > 0$.

(3 points)

Problem 4. What is the moment of inertia of the cylinder from Problem 3 about an axis parallel to the axis of symmetry, and tangential to the outer side wall?

(1 point)

Problem 5. A yo-yo is made of two identical cylinders of radius R attached at their centers to a cylindrical axle with radius $r < R$. All three cylinders have the same height and are made of the same material (with constant density of mass). A string is attached to and wrapped around the axle. Holding the free end of the string firm, we release the yo-yo.



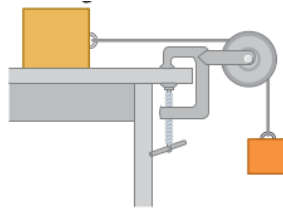
Find the acceleration of the center of mass of the yo-yo. Assume that the string does not slip while unwinding.

(4 points)

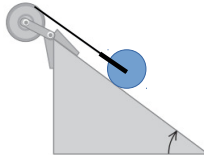
Problem 6. A uniform solid ball and a ring start rolling without slipping from the top of an inclined plane with slope angle α . The initial velocity of the ball is zero. What should the initial linear velocity of the ring be, so that both objects travel the same distance in the same time t ?

(4 points)

- Problem 7.** A block with mass m_1 resting on a horizontal, frictionless surface is attached to another block with mass m_2 by a thin, light wire that passes over a pulley. The pulley has the shape of a uniform disk with mass M and radius R . After the system is released, find (a) the tension in the wire on both sides of the pulley, (b) the acceleration of the box, and (c) the horizontal and vertical components of the force that the axle exerts on the pulley.
(2 + 2 + 1 points)



- Problem 8.** A uniform sphere with mass $5M$ and radius R is placed on an plane inclined at an angle α . A string is attached by a yoke to a frictionless axle through the center of the sphere so that the sphere can rotate about the axle. The string is wrapped around a flywheel in the shape of a disk with radius R and mass M . The sphere rolls without slipping. What is the acceleration of the center of mass of the sphere?
(5 points)



- Problem 9.** A block with mass m is attached to one end of a light rope that runs over a pulley with the moment of inertia I and radius R . The other end of the rope is attached to a spring with spring constant k . Find the period of oscillations of this system.
(5 points)

